mcc2024-benchmark-description

March 2024

1 Description

The benchmark instances are synthetic and come from the recent work on pseudo-boolean model counting [1]. The CNF instances provided are original pseudo-boolean formulas and have been converted into CNFs using the pbencoder binary from the PBLib[1] repository with default settings. The CNFs are projected counting instances, whereby the projection set is the set of original PB variables.

The instances comes from 2 different settings of synthetic benchmarks as follows:

Auction - 1000 instances The benchmarks are inspired by auction settings. We have n items and m participants, each with a minimum utility that we want to reach. We count the number of ways we can allocate the items such that all participants have utility at least their minimum threshold. We allow each item to be shared by multiple participants.

Knapsack - 1000 instances Multi-dimension knapsack type problems, where there are n items, and constraints on different dimensions. We want to count the number of ways of selecting items such that the constraints on each dimension is not exceeded. Some items do not have certain dimensions.

For an idea of how difficult the instances are, please refer to our paper[2] where we ran experiments with different CNF model counters using 3600s timeout. We also provide the counts and runtime of GPMC and D4[3] in the spreadsheets.

References

[1] Suwei Yang and Kuldeep S. Meel. Engineering an exact pseudo-boolean model counter. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence*, volume 38, pages 8200–8208, 2024.

[1] https://github.com/master-keying/pblib
[2] https://arxiv.org/abs/2312.12341
[3] We used the mcc2022 binary